A11 Box - Parts to Buy

United Control Box, wiring rev 0.7 (direct wired; no terminal blocks, only the L / N busbars)
Enclosure: bought 800 × 1000 × 250, swing door

Already have: breakers, EMI filter, contactor, VFD, PSUs, drives (with ALM), E-stop (1 NC + block below), ON/OFF, connectors, NPN NO sensors, fan, Schuko, earth bar.

	BOUGHT		
☑	Enclosure 800 W × 1000 H × 250 D mm, steel, swing door hinged on the left, mounting plate included, no removable panels	x1	wall mount
☑	Relay Finder 55.34.9.024.0000 + socket (RXZE2S114M look-alike), 24 V DC (K1: pole 1 latch, 2 spare, 3 empty gap, 4 → KM1 coil; AC15 350 VA ≥ 200 VA coil inrush)	x1	in socket
☑	Fuse holder RT18-32 (or RT14) 1P, DIN rail, for 10 × 38 mm fuse links (F1, F2)	x2	DIN
☑	Fuse link 10 × 38 mm (RO15) gG 2 A: use these for F1 and F2	x2 + spares	-
☑	Fuse link 10 × 38 mm aM 2 A: do NOT use for F1 / F2 (aM = motor short-circuit only, no overload protection)	-	spare / return
☑	230 V busbar 160 × 16 × 42 mm, 12 points, on insulators: XL (switched L) and XN (all N); vertical, right of the right vertical duct	x2	screws to plate
☑	Earth bar, copper 150 × 25 mm on two 17 mm red insulators, 4 points: PE 1 and PE 2, linked by wire 138. Up to 2 conductors per point (owner decision; IEC 60204-1 asks for one protective conductor per point - a 12-point bar removes the deviation)	x2	bottom wall
☑	Slotted wire duct 40 W × 60 H with cover, 2 m (cut list in A11_Enclosure_Selection.md; motor lane VM 230 mm)	x3	screws to plate
	ENCLOSURE & MOUNTING		
☑	DIN rail TS35 × 7.5, slotted: one 330 mm piece (row A, right of the plate-mounted EMI filter), + 4 end stops	0.5 m + x4	screws to plate
☐	DIN rail earth clamp, M6 stud	x1	DIN
☐	Door earth strap, braided ≥ 6 mm², 2 m, M6 lugs (door stud → earth bar, through the hinge loop)	x1	-
☑	M6 earth stud kit (bolt, 2 nuts, serrated washers): body (wire 139) and plate (wire 141), only where the enclosure has none	x2	body / plate
☑	Cable gland M20, nylon, IP68 (mains + pump cord)	x2	bottom wall
☑	80 mm plastic filter adapters, flanged: one on the 24 V fan (left wall), one on the outlet (right wall); 80 × 80 cut-outs	x2	side walls
	EMC (spindle cable W1 and stepper motor cables)		
☐	EMC shield clamp (saddle) for Ø 8–12 mm, screwed to the plate: W1 at the power lane (wire 127)	x1	plate
☐	HDC-HA metal hood + housing with EMC cable gland (360° shield), both halves; + brass EMC gland at the spindle terminal box	x1 set + x1	bottom wall / spindle
☐	Shield clamps for the motor cables at the drives (LiYCY braid to the plate)	x3	plate
	CONTROL PARTS		
☑	RichAuto A11E kit (handheld + board + 50-pin cable) - 24 V DC 3 A	x1	standoffs
☑	22 mm LED pilot lamp 24 V DC - yellow / green / red	x1 each	door
☐	NC contact block Schneider ZB2-BE102 (or the clone's 1 NC block) clipped onto the E-stop base: NC2 → A11 X7	x1	door
☐	GX16-5 plug with pins 3-4 bridged: dummy plug when the tool setter is unplugged (over-travel is in the X7 loop)	x1	XT
	WIRE - colours IEC 60204-1 (3D routing, 121 conductors, +10 % +15 cm each, rounded up); stock bought 2026-10-27; per-wire marking in A11_Wire_List.pdf		
☑	1 mm² H05V-K yellow, 90 m: every control, signal and 230 V branch wire; function colour as a band at both ends (black L / 48 V, blue N, red KM1 coil, dark blue +24 V / signals, blue + white 0 V); 5V / PULSE / DIR twisted in threes. Needs 67 m	90 m	
☑	2.5 mm² H07V-K black (mains L 101, VFD L 115): needs 3 m	4 m	
☑	2.5 mm² H07V-K blue (mains N 102, VFD N 116): needs 3 m	4 m	
☑	2.5 mm² H07V-K brown + dark-blue band (+24 V PSU → F1 / F2, 162 / 164): needs 3 m	4 m	
☑	2.5 mm² H07V-K green/yellow (PE): needs 9 m	4 m	
☐	2.5 mm² (or 1.5 mm²) H07V-K green/yellow: the rest of the PE wires (stock check in A11_Wire_List.pdf)	5 m	
☑	4 mm² H07V-K black (L, supply path behind QF0): needs 4 m	5 m	
☑	4 mm² H07V-K blue (N, supply path behind QF0): needs 2 m	5 m	
☑	4 mm² H07V-K green/yellow (PE): needs 4 m	5 m	
☑	4 mm² H07V-K brown: not used (spare)	5 m	
☐	Coloured sleeves or tape for the 1 mm² bands: black, blue, red, dark blue, white (with blue for 0 V)	1 set	
☐	Control cable LiYCY 4 × 1.0 shielded (drive → motor connector, 1 per axis; also for the motor side)	4 m	motor lane
☐	Shielded cable 4G1.5, e.g. ÖLFLEX CLASSIC 110 CY (W1: VFD → spindle connector)	2 m	power lane
	FINISHING & TOOLS		
☑	Insulated ferrules: 0.5 ×100, 1.0 ×50, 1.5 ×100, 2.5 ×50, 4 ×50	1 box each	
☐	1 mm² ferrules: the yellow wire needs more than the box of 50 (count in A11_Wire_List.pdf)	1 box ×100	
☑	Twin ferrules: 2 × 0.5 ×50, 2 × 1.5 ×10, 2 × 2.5 ×10, 2 × 4 ×10 (EMI filter LOAD L: 105 + 111)	1 box each	
☐	Twin ferrules 2 × 1 mm² (the 2 × 0.5 box no longer fits the 1 mm² wire)	1 box ×50	
☑	Wire number markers 0-9 for 0.5-4 mm²; spiral wrap 10 mm (door hinge loop)	1 set + 1 m	
☑	Knock-out punches / step drill: 16.2 (GX16), 20.5 (M20), 22.5 (door devices); nibbler or jigsaw for 2 × 80 × 80 (fan, outlet)	if needed	bottom wall / door
	DECIDE BEFORE ORDERING		
☐	C14 open: FC300 input ≈ 22–24 A at full load (11 A is output). Z1 fixed at 20 A (no larger filter): cap the spindle at ≈ 1.5 kW in the VFD (≈ 15 A input), check with a clamp meter. Z1 LOAD L: 2 × 4 mm²?	-	
☐	48 V drive feeds 150-155 are 1 mm² (10.4 A) on the 15 A PSU: fuse each drive (3 × 6.3 A + holders) or wire them in 1.5-2.5 mm²	x3 or 6 m	

Lengths: 3D duct routes ±15-20 %, Verification/A11_Wire_Lengths_rev07.csv; regenerate with make_parts_list.py.